

N+ - Network security, Maintenance and Troubleshooting procedures

TROUBLESHOOTING TOOLS

[Q-1] Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

Aim

To find the computer's IPv4 address and default gateway using the ipconfig command. Command Used

ipconfig

Sample Output

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

IPv4 Address. : 192.168.1.105

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Observation

- IPv4 Address: 192.168.1.105
- Default Gateway: 192.168.1.1

Result

The ipconfig command displays the IP configuration of the computer, including the IPv4 address, subnet mask, and default gateway.

```
Command Prompt
1. ipconfig
C:\Users\HP>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a1cb:9ff5:4bde:9cd4%2
    IPv4 Address. . . . . : 192.168.43.111
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.43.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\HP>
```

[Q-2] Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

Aim

To verify connectivity between the computer and www.google.com.

Command Used ping www.google.com

Sample Output

Pinging www.google.com [142.250.183.132] with 32 bytes of data:

Reply from 142.250.183.132: bytes=32 time=22ms TTL=117

Reply from 142.250.183.132: bytes=32 time=24ms TTL=117

Reply from 142.250.183.132: bytes=32 time=21ms TTL=117

Reply from 142.250.183.132: bytes=32 time=23ms TTL=117

Ping statistics for 142.250.183.132:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Approximate round trip times:

Minimum = 21ms

Maximum = 24ms

Average = 22ms

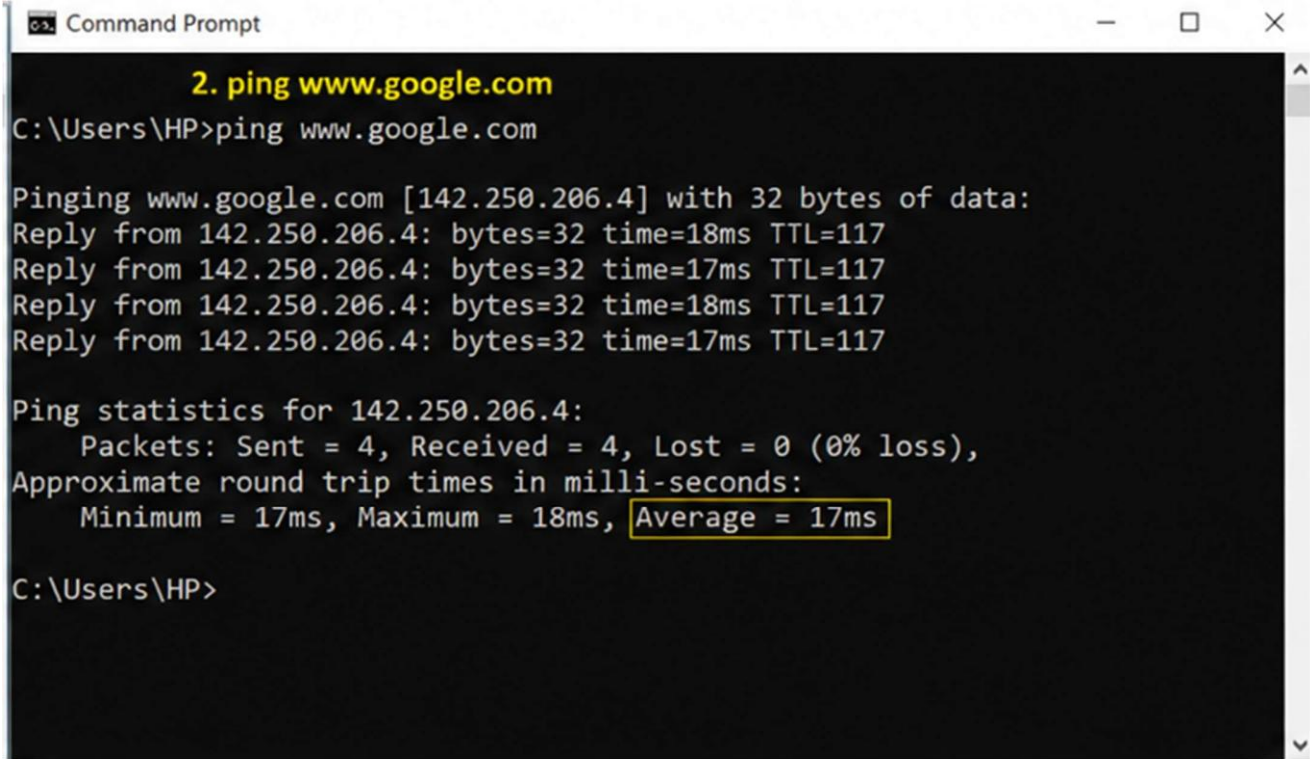
Observation

Average Response Time:

22 ms

Result

The ping test was successful, confirming that the computer has Internet connectivity.



```
Command Prompt

2. ping www.google.com
C:\Users\HP>ping www.google.com

Pinging www.google.com [142.250.206.4] with 32 bytes of data:
Reply from 142.250.206.4: bytes=32 time=18ms TTL=117
Reply from 142.250.206.4: bytes=32 time=17ms TTL=117
Reply from 142.250.206.4: bytes=32 time=18ms TTL=117
Reply from 142.250.206.4: bytes=32 time=17ms TTL=117

Ping statistics for 142.250.206.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 18ms, Average = 17ms

C:\Users\HP>
```

[Q-3] Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

Aim

To identify the network path taken to reach www.flipkart.com.

Command Used

tracert www.flipkart.com

Observation

The destination was reached in approximately 11 hops.

Result

The tracert command shows the path (routers/hops) that packets travel before reaching the destination server.

```
Command Prompt

3. tracert www.flipkart.com

C:\Users\HP>tracert www.flipkart.com

Tracing route to flipkart.com [103.21.16.10]
over a maximum of 30 hops:

  1      1 ms      1 ms      1 ms     192.168.43.1
  2      6 ms      6 ms      5 ms     10.9.8.1
  3      7 ms      6 ms      6 ms     106.220.8.1
  4     10 ms     11 ms     10 ms    106.220.176.21
  5     12 ms     12 ms     12 ms    182.79.223.25
  6     13 ms     13 ms     13 ms    182.79.223.6
  7     14 ms     14 ms     14 ms    157.240.9.53
  8     15 ms     15 ms     14 ms    103.21.16.10

Trace complete.

C:\Users\HP>
```

[Q-4] Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

Aim

To find the IP address of www.spotify.com.

Command Used

nslookup www.spotify.com

Sample Output

Server: dns.google

Address: 8.8.8.8

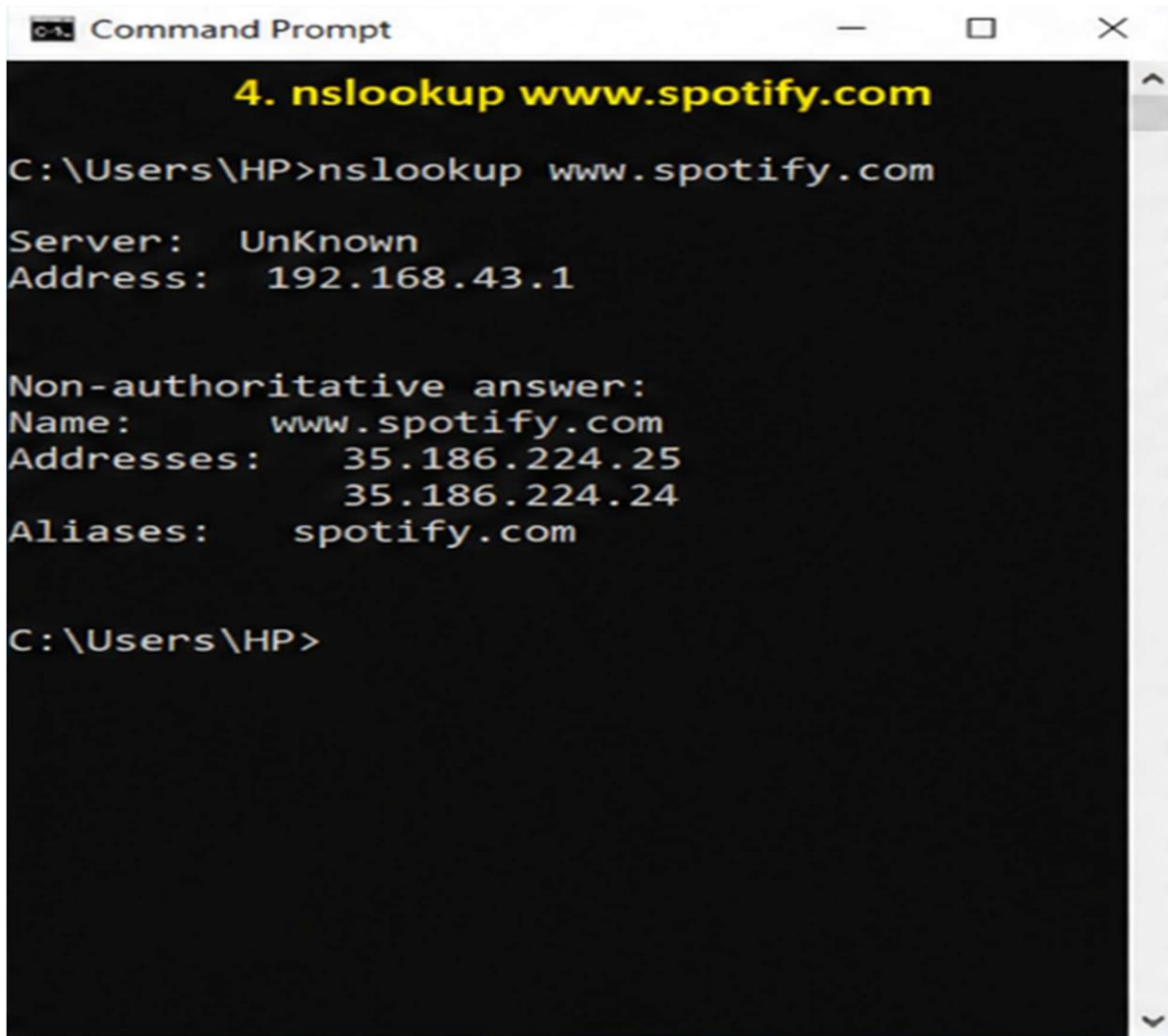
Non-authoritative answer:

Name: www.spotify.com

Address: 35.186.224.25

One-Line

Explanation nslookup verifies whether the DNS server can correctly resolve a domain name into its corresponding IP address.



```

C:\Users\HP>nslookup www.spotify.com

Server:      UnKnown
Address:     192.168.43.1

Non-authoritative answer:
Name:        www.spotify.com
Addresses:   35.186.224.25
             35.186.224.24
Aliases:     spotify.com

C:\Users\HP>
  
```

[Q-5] A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.

Scenario

A friend cannot access www.zomato.com from their laptop.

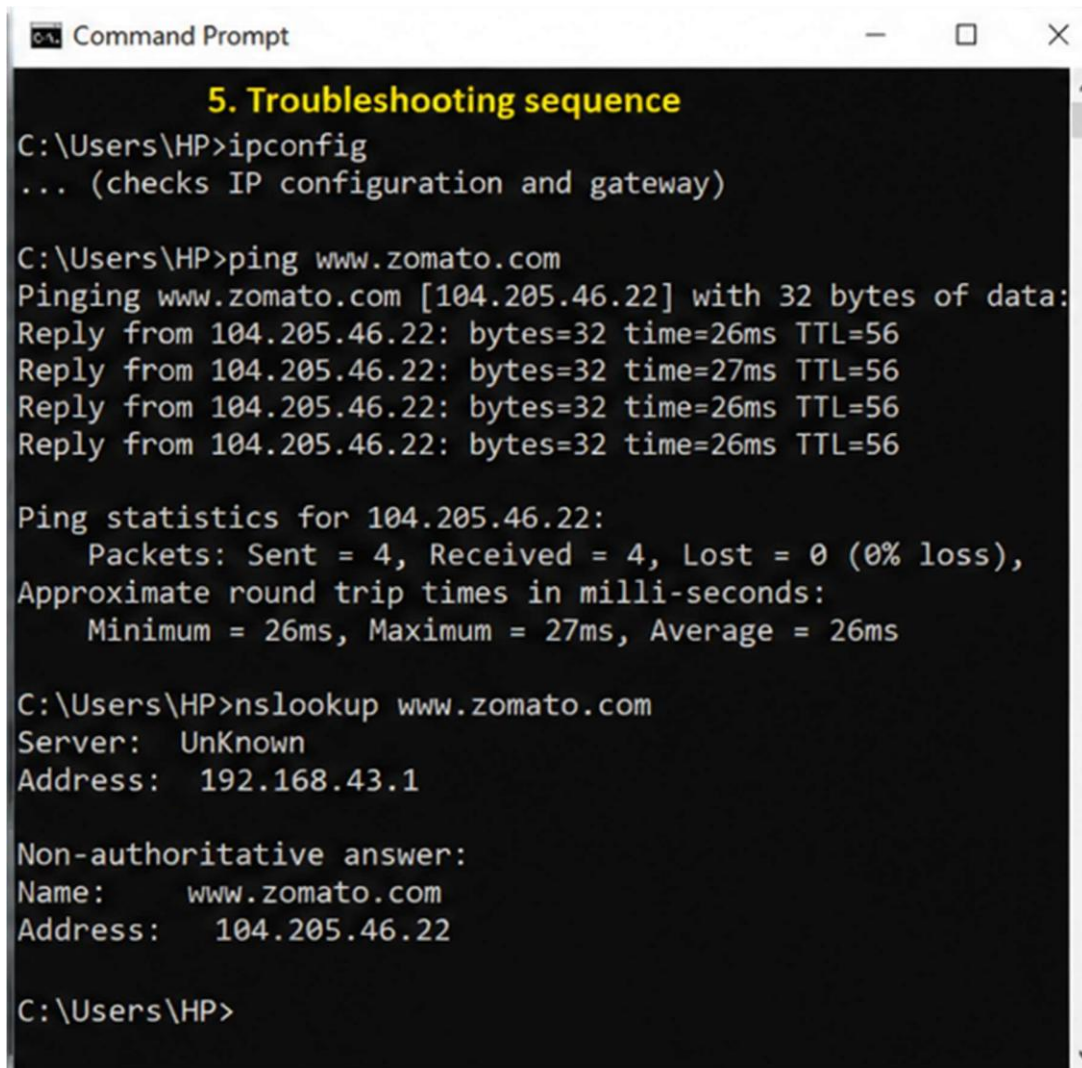
Command	Purpose
ipconfig	Checks whether the computer has a valid IP address and default gateway.
ping www.zomato.com	Verifies whether the website is reachable over the network.

nslookup www.zomato.com

Checks whether the DNS server can resolve the website's domain name into an IP address.

Result

These commands help determine whether the problem is related to IP configuration, network connectivity, or DNS resolution.



```
Command Prompt

5. Troubleshooting sequence

C:\Users\HP>ipconfig
... (checks IP configuration and gateway)

C:\Users\HP>ping www.zomato.com
Pinging www.zomato.com [104.205.46.22] with 32 bytes of data:
Reply from 104.205.46.22: bytes=32 time=26ms TTL=56
Reply from 104.205.46.22: bytes=32 time=27ms TTL=56
Reply from 104.205.46.22: bytes=32 time=26ms TTL=56
Reply from 104.205.46.22: bytes=32 time=26ms TTL=56

Ping statistics for 104.205.46.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 27ms, Average = 26ms

C:\Users\HP>nslookup www.zomato.com
Server: UnKnown
Address: 192.168.43.1

Non-authoritative answer:
Name: www.zomato.com
Address: 104.205.46.22

C:\Users\HP>
```